ONTAP NAS drivers

Astra Trident

NetApp
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Configure an ONTAP NAS backend

Learn about configuring an ONTAP backend with ONTAP and Cloud Volumes ONTAP NAS drivers.

- Preparation
- Configuration and examples

> Astra Control provides seamless protection, disaster recovery, and mobility (moving volumes between Kubernetes clusters) for volumes created with the `ontap-nas`, `ontap-nas-flexgroup`, and `ontap-san` drivers. See Astra Control replication prerequisites for details.
>
> - You must use `ontap-nas` for production workloads that require data protection, disaster recovery, and mobility.
> - Use `ontap-san-economy` when anticipated volume usage is expected to be much higher than what ONTAP supports.
> - Use `ontap-nas-economy` only where anticipated volume usage is expected to be much higher than what ONTAP supports, and the `ontap-san-economy` driver cannot be used.
> - Do not use use `ontap-nas-economy` if you anticipate the need for data protection, disaster recovery, or mobility.

User permissions

Astra Trident expects to be run as either an ONTAP or SVM administrator, typically using the `admin` cluster user or a `vsadmin` SVM user, or a user with a different name that has the same role. For Amazon FSx for NetApp ONTAP deployments, Astra Trident expects to be run as either an ONTAP or SVM administrator, using the cluster `fsxadmin` user or a `vsadmin` SVM user, or a user with a different name that has the same role. The `fsxadmin` user is a limited replacement for the cluster admin user.

> If you use the `limitAggregateUsage` parameter, cluster admin permissions are required. When using Amazon FSx for NetApp ONTAP with Astra Trident, the `limitAggregateUsage` parameter will not work with the `vsadmin` and `fsxadmin` user accounts. The configuration operation will fail if you specify this parameter.

While it is possible to create a more restrictive role within ONTAP that a Trident driver can use, we don't recommend it. Most new releases of Trident will call additional APIs that would have to be accounted for, making upgrades difficult and error-prone.

Prepare to configure a backend with ONTAP NAS drivers

Learn about how to prepare to configure an ONTAP backend with ONTAP NAS drivers. For all ONTAP backends, Astra Trident requires at least one aggregate assigned to the SVM.

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Remember that you can also run more than one driver, and create storage classes that point to one or the other. For example, you could configure a Gold class that uses the `ontap-nas` driver and a Bronze class that uses the `ontap-nas-economy` one.

All your Kubernetes worker nodes must have the appropriate NFS tools installed. See here for more details.

Authentication

Astra Trident offers two modes of authenticating an ONTAP backend.

- Credential-based: The username and password to an ONTAP user with the required permissions. It is recommended to use a pre-defined security login role, such as `admin` or `vsadmin` to ensure maximum compatibility with ONTAP versions.
- Certificate-based: Astra Trident can also communicate with an ONTAP cluster using a certificate installed on the backend. Here, the backend definition must contain Base64-encoded values of the client certificate, key, and the trusted CA certificate if used (recommended).

You can update existing backends to move between credential-based and certificate-based methods. However, only one authentication method is supported at a time. To switch to a different authentication method, you must remove the existing method from the backend configuration.

> If you attempt to provide **both credentials and certificates**, backend creation will fail with an error that more than one authentication method was provided in the configuration file.

Enable credential-based authentication

Astra Trident requires the credentials to an SVM-scoped/cluster-scoped admin to communicate with the ONTAP backend. It is recommended to make use of standard, pre-defined roles such as `admin` or `vsadmin`. This ensures forward compatibility with future ONTAP releases that might expose feature APIs to be used by future Astra Trident releases. A custom security login role can be created and used with Astra Trident, but is not recommended.

A sample backend definition will look like this:

YAML

```yaml
---
version: 1
backendName: ExampleBackend
storageDriverName: ontap-nas
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
svm: svm_nfs
username: vsadmin
password: password
```

JSON

```json
{
  "version": 1,
  "backendName": "ExampleBackend",
  "storageDriverName": "ontap-nas",
  "managementLIF": "10.0.0.1",
  "dataLIF": "10.0.0.2",
  "svm": "svm_nfs",
  "username": "vsadmin",
  "password": "password"
}
```

Keep in mind that the backend definition is the only place the credentials are stored in plain text. After the backend is created, usernames/passwords are encoded with Base64 and stored as Kubernetes secrets. The creation/updation of a backend is the only step that requires knowledge of the credentials. As such, it is an admin-only operation, to be performed by the Kubernetes/storage administrator.

Enable certificate-based Authentication

New and existing backends can use a certificate and communicate with the ONTAP backend. Three parameters are required in the backend definition.

- clientCertificate: Base64-encoded value of client certificate.
- clientPrivateKey: Base64-encoded value of associated private key.
- trustedCACertificate: Base64-encoded value of trusted CA certificate. If using a trusted CA, this parameter must be provided. This can be ignored if no trusted CA is used.

A typical workflow involves the following steps.

Steps

1. Generate a client certificate and key. When generating, set Common Name (CN) to the ONTAP user to authenticate as.

```
openssl req -x509 -nodes -days 1095 -newkey rsa:2048 -keyout k8senv.key
-out k8senv.pem -subj "/C=US/ST=NC/L=RTP/O=NetApp/CN=vsadmin"
```

2. Add trusted CA certificate to the ONTAP cluster. This might be already handled by the storage administrator. Ignore if no trusted CA is used.

```
security certificate install -type server -cert-name <trusted-ca-cert-
name> -vserver <vserver-name>
ssl modify -vserver <vserver-name> -server-enabled true -client-enabled
true -common-name <common-name> -serial <SN-from-trusted-CA-cert> -ca
<cert-authority>
```

3. Install the client certificate and key (from step 1) on the ONTAP cluster.

```
security certificate install -type client-ca -cert-name <certificate-
name> -vserver <vserver-name>
security ssl modify -vserver <vserver-name> -client-enabled true
```

4. Confirm the ONTAP security login role supports `cert` authentication method.

```
security login create -user-or-group-name vsadmin -application ontapi
-authentication-method cert -vserver <vserver-name>
security login create -user-or-group-name vsadmin -application http
-authentication-method cert -vserver <vserver-name>
```

5. Test authentication using certificate generated. Replace <ONTAP Management LIF> and <vserver name> with Management LIF IP and SVM name. You must ensure the LIF has its service policy set to `default-data-management`.

```
curl -X POST -Lk https://<ONTAP-Management-
LIF>/servlets/netapp.servlets.admin.XMLrequest_filer --key k8senv.key
--cert ~/k8senv.pem -d '<?xml version="1.0" encoding="UTF-8"?><netapp
xmlns="http://www.netapp.com/filer/admin" version="1.21"
vfiler="<vserver-name>"><vserver-get></vserver-get></netapp>'
```

6. Encode certificate, key and trusted CA certificate with Base64.

```
base64 -w 0 k8senv.pem >> cert_base64
base64 -w 0 k8senv.key >> key_base64
base64 -w 0 trustedca.pem >> trustedca_base64
```

7. Create backend using the values obtained from the previous step.

```
cat cert-backend-updated.json
{
"version": 1,
"storageDriverName": "ontap-nas",
"backendName": "NasBackend",
"managementLIF": "1.2.3.4",
"dataLIF": "1.2.3.8",
"svm": "vserver_test",
"clientCertificate": "Faaaakkkkeeee...Vaaalllluuuueeee",
"clientPrivateKey": "LS0tFaKE...0VaLuES0tLS0K",
"storagePrefix": "myPrefix_"
}

#Update backend with tridentctl
tridentctl update backend NasBackend -f cert-backend-updated.json -n
trident
+------------+----------------+--------------------------------------
+--------+---------+
|    NAME    | STORAGE DRIVER |                  UUID                |
STATE  | VOLUMES |
+------------+----------------+--------------------------------------
+--------+---------+
| NasBackend | ontap-nas      | 98e19b74-aec7-4a3d-8dcf-128e5033b214 |
online |       9 |
+------------+----------------+--------------------------------------
+--------+---------+
```

Update authentication methods or rotate credentials

You can update an existing backend to use a different authentication method or to rotate their credentials. This works both ways: backends that make use of username/password can be updated to use certificates; backends that utilize certificates can be updated to username/password based. To do this, you must remove the existing authentication method and add the new authentication method. Then use the updated backend.json file containing the required parameters to execute `tridentctl update backend`.

```
cat cert-backend-updated.json
{
"version": 1,
"storageDriverName": "ontap-nas",
"backendName": "NasBackend",
"managementLIF": "1.2.3.4",
"dataLIF": "1.2.3.8",
"svm": "vserver_test",
"username": "vsadmin",
"password": "password",
"storagePrefix": "myPrefix_"
}

#Update backend with tridentctl
tridentctl update backend NasBackend -f cert-backend-updated.json -n
trident
+------------+----------------+--------------------------------------
+--------+---------+
|    NAME    | STORAGE DRIVER |                 UUID                 |
STATE  | VOLUMES |
+------------+----------------+--------------------------------------
+--------+---------+
| NasBackend | ontap-nas      | 98e19b74-aec7-4a3d-8dcf-128e5033b214 |
online |       9 |
+------------+----------------+--------------------------------------
+--------+---------+
```

> When rotating passwords, the storage administrator must first update the password for the user on ONTAP. This is followed by a backend update. When rotating certificates, multiple certificates can be added to the user. The backend is then updated to use the new certificate, following which the old certificate can be deleted from the ONTAP cluster.

Updating a backend does not disrupt access to volumes that have already been created, nor impact volume connections made after. A successful backend update indicates that Astra Trident can communicate with the ONTAP backend and handle future volume operations.

Manage NFS export policies

Astra Trident uses NFS export policies to control access to the volumes that it provisions.

Astra Trident provides two options when working with export policies:

- Astra Trident can dynamically manage the export policy itself; in this mode of operation, the storage administrator specifies a list of CIDR blocks that represent admissible IP addresses. Astra Trident adds node IPs that fall in these ranges to the export policy automatically. Alternatively, when no CIDRs are specified, any global-scoped unicast IP found on the nodes will be added to the export policy.
- Storage administrators can create an export policy and add rules manually. Astra Trident uses the default

export policy unless a different export policy name is specified in the configuration.

Dynamically manage export policies

The 20.04 release of CSI Trident provides the ability to dynamically manage export policies for ONTAP backends. This provides the storage administrator the ability to specify a permissible address space for worker node IPs, rather than defining explicit rules manually. It greatly simplifies export policy management; modifications to the export policy no longer require manual intervention on the storage cluster. Moreover, this helps restrict access to the storage cluster only to worker nodes that have IPs in the range specified, supporting a fine-grained and automated management.

> The dynamic management of export policies is only available for CSI Trident. It is important to ensure that the worker nodes are not being NATed.

Example

There are two configuration options that must be used. Here's an example backend definition:

```
---
version: 1
storageDriverName: ontap-nas
backendName: ontap_nas_auto_export
managementLIF: 192.168.0.135
svm: svm1
username: vsadmin
password: password
autoExportCIDRs:
- 192.168.0.0/24
autoExportPolicy: true
```

> When using this feature, you must ensure that the root junction in your SVM has a previously created export policy with an export rule that permits the node CIDR block (such as the default export policy). Always follow NetApp's recommended best practice of dedicating a SVM for Astra Trident.

Here is an explanation of how this feature works using the example above:

- `autoExportPolicy` is set to `true`. This indicates that Astra Trident will create an export policy for the `svm1` SVM and handle the addition and deletion of rules using `autoExportCIDRs` address blocks. For example, a backend with UUID 403b5326-8482-40db-96d0-d83fb3f4daec and `autoExportPolicy` set to `true` creates an export policy named `trident-403b5326-8482-40db-96d0-d83fb3f4daec` on the SVM.
- `autoExportCIDRs` contains a list of address blocks. This field is optional and it defaults to ["0.0.0.0/0", "::/0"]. If not defined, Astra Trident adds all globally-scoped unicast addresses found on the worker nodes.

In this example, the `192.168.0.0/24` address space is provided. This indicates that Kubernetes node IPs that fall within this address range will be added to the export policy that Astra Trident creates. When Astra Trident registers a node it runs on, it retrieves the IP addresses of the node and checks them against the address blocks provided in `autoExportCIDRs`. After filtering the IPs, Astra Trident creates export policy rules

for the client IPs it discovers, with one rule for each node it identifies.

You can update `autoExportPolicy` and `autoExportCIDRs` for backends after you create them. You can append new CIDRs for a backend that is automatically managed or delete existing CIDRs. Exercise care when deleting CIDRs to ensure that existing connections are not dropped. You can also choose to disable `autoExportPolicy` for a backend and fall back to a manually created export policy. This will require setting the `exportPolicy` parameter in your backend config.

After Astra Trident creates or updates a backend, you can check the backend using `tridentctl` or the corresponding `tridentbackend` CRD:

```
./tridentctl get backends ontap_nas_auto_export -n trident -o yaml
items:
- backendUUID: 403b5326-8482-40db-96d0-d83fb3f4daec
  config:
    aggregate: ""
    autoExportCIDRs:
    - 192.168.0.0/24
    autoExportPolicy: true
    backendName: ontap_nas_auto_export
    chapInitiatorSecret: ""
    chapTargetInitiatorSecret: ""
    chapTargetUsername: ""
    chapUsername: ""
    dataLIF: 192.168.0.135
    debug: false
    debugTraceFlags: null
    defaults:
      encryption: "false"
      exportPolicy: <automatic>
      fileSystemType: ext4
```

As nodes are added to a Kubernetes cluster and registered with the Astra Trident controller, export policies of existing backends are updated (provided they fall in the address range specified in `autoExportCIDRs` for the backend).

When a node is removed, Astra Trident checks all backends that are online to remove the access rule for the node. By removing this node IP from the export policies of managed backends, Astra Trident prevents rogue mounts, unless this IP is reused by a new node in the cluster.

For previously existing backends, updating the backend with `tridentctl update backend` will ensure that Astra Trident manages the export policies automatically. This will create a new export policy named after the backend's UUID and volumes that are present on the backend will use the newly created export policy when they are mounted again.

> Deleting a backend with auto-managed export policies will delete the dynamically created export policy. If the backend is re-created, it is treated as a new backend and will result in the creation of a new export policy.

If the IP address of a live node is updated, you must restart the Astra Trident pod on the node. Astra Trident will then update the export policy for backends it manages to reflect this IP change.

ONTAP NAS configuration options and examples

Learn about how to create and use ONTAP NAS drivers with your Astra Trident installation. This section provides backend configuration examples and details about how to map backends to StorageClasses.

Backend configuration options

See the following table for the backend configuration options:

Parameter	Description	Default
`version`		Always 1
`storageDriverName`	Name of the storage driver	“ontap-nas”, “ontap-nas-economy”, “ontap-nas-flexgroup”, “ontap-san”, “ontap-san-economy”
`backendName`	Custom name or the storage backend	Driver name + “_” + dataLIF
`managementLIF`	IP address of a cluster or SVM management LIF For seamless MetroCluster switchover, you must specify an SVM management LIF. A fully-qualified domain name (FQDN) can be specified. Can be set to use IPv6 addresses if Astra Trident was installed using the `--use-ipv6` flag. IPv6 addresses must be defined in square brackets, such as [28e8:d9fb:a825:b7bf:69a8:d02f:9e7b:3555].	“10.0.0.1”, “[2001:1234:abcd::fefe]”

Parameter	Description	Default
`dataLIF`	IP address of protocol LIF. We recommend specifying `dataLIF`. If not provided, Astra Trident fetches data LIFs from the SVM. You can specify a fully-qualified domain name (FQDN) to be used for the NFS mount operations, allowing you to create a round-robin DNS to load-balance across multiple data LIFs. Can be changed after initial setting. Refer to Update `dataLIF` after initial configuration. Can be set to use IPv6 addresses if Astra Trident was installed using the `--use-ipv6` flag. IPv6 addresses must be defined in square brackets, such as [28e8:d9fb:a825:b7bf:69a8:d02f:9e7b:3555].	Specified address or derived from SVM, if not specified (not recommended)
`autoExportPolicy`	Enable automatic export policy creation and updating [Boolean]. Using the `autoExportPolicy` and `autoExportCIDRs` options, Astra Trident can manage export policies automatically.	false
`autoExportCIDRs`	List of CIDRs to filter Kubernetes' node IPs against when `autoExportPolicy` is enabled. Using the `autoExportPolicy` and `autoExportCIDRs` options, Astra Trident can manage export policies automatically.	["0.0.0.0/0", "::/0"]`
`labels`	Set of arbitrary JSON-formatted labels to apply on volumes	""
`clientCertificate`	Base64-encoded value of client certificate. Used for certificate-based auth	""
`clientPrivateKey`	Base64-encoded value of client private key. Used for certificate-based auth	""

Parameter	Description	Default
`trustedCACertificate`	Base64-encoded value of trusted CA certificate. Optional. Used for certificate-based auth	“”
`username`	Username to connect to the cluster/SVM. Used for credential-based auth	
`password`	Password to connect to the cluster/SVM. Used for credential-based auth	
`svm`	Storage virtual machine to use	Derived if an SVM `managementLIF` is specified
`storagePrefix`	Prefix used when provisioning new volumes in the SVM. Cannot be updated after you set it	“trident”
`limitAggregateUsage`	Fail provisioning if usage is above this percentage. **Does not apply to Amazon FSx for ONTAP**	“” (not enforced by default)
`limitVolumeSize`	Fail provisioning if requested volume size is above this value.	“” (not enforced by default)
`limitVolumeSize`	Fail provisioning if requested volume size is above this value. Also restricts the maximum size of the volumes it manages for qtrees and LUNs, and the `qtreesPerFlexvol` option allows customizing the maximum number of qtrees per FlexVol.	“” (not enforced by default)
`lunsPerFlexvol`	Maximum LUNs per Flexvol, must be in range [50, 200]	“100”
`debugTraceFlags`	Debug flags to use when troubleshooting. Example, {“api”:false, “method”:true} Do not use `debugTraceFlags` unless you are troubleshooting and require a detailed log dump.	null

Parameter	Description	Default
`nfsMountOptions`	Comma-separated list of NFS mount options. The mount options for Kubernetes-persistent volumes are normally specified in storage classes, but if no mount options are specified in a storage class, Astra Trident will fall back to using the mount options specified in the storage backend's configuration file. If no mount options are specified in the storage class or the configuration file, Astra Trident will not set any mount options on an associated persistent volume.	“”
`qtreesPerFlexvol`	Maximum Qtrees per FlexVol, must be in range [50, 300]	“200”
`useREST`	Boolean parameter to use ONTAP REST APIs. **Tech preview** `useREST` is provided as a **tech preview** that is recommended for test environments and not for production workloads. When set to `true`, Astra Trident will use ONTAP REST APIs to communicate with the backend. This feature requires ONTAP 9.11.1 and later. In addition, the ONTAP login role used must have access to the `ontap` application. This is satisfied by the pre-defined `vsadmin` and `cluster-admin` roles. `useREST` is not supported with MetroCluster.	false

Backend configuration options for provisioning volumes

You can control default provisioning using these options in the `defaults` section of the configuration. For an example, see the configuration examples below.

Parameter	Description	Default
`spaceAllocation`	Space-allocation for LUNs	“true”
`spaceReserve`	Space reservation mode; “none” (thin) or “volume” (thick)	“none”

Parameter	Description	Default
`snapshotPolicy`	Snapshot policy to use	“none”
`qosPolicy`	QoS policy group to assign for volumes created. Choose one of qosPolicy or adaptiveQosPolicy per storage pool/backend	“”
`adaptiveQosPolicy`	Adaptive QoS policy group to assign for volumes created. Choose one of qosPolicy or adaptiveQosPolicy per storage pool/backend. Not supported by ontap-nas-economy.	“”
`snapshotReserve`	Percentage of volume reserved for snapshots “0”	If `snapshotPolicy` is “none”, else “”
`splitOnClone`	Split a clone from its parent upon creation	“false”
`encryption`	Enable NetApp Volume Encryption (NVE) on the new volume; defaults to `false`. NVE must be licensed and enabled on the cluster to use this option. If NAE is enabled on the backend, any volume provisioned in Astra Trident will be NAE enabled. For more information, refer to: How Astra Trident works with NVE and NAE.	“false”
`tieringPolicy`	Tiering policy to use “none”	“snapshot-only” for pre-ONTAP 9.5 SVM-DR configuration
`unixPermissions`	Mode for new volumes	“777” for NFS volumes; empty (not applicable) for SMB volumes
`snapshotDir`	Controls visibility of the `.snapshot` directory	“false”
`exportPolicy`	Export policy to use	“default”
`securityStyle`	Security style for new volumes. NFS supports `mixed` and `unix` security styles. SMB supports `mixed` and `ntfs` security styles.	NFS default is `unix`. SMB default is `ntfs`.

Using QoS policy groups with Astra Trident requires ONTAP 9.8 or later. It is recommended to use a non-shared QoS policy group and ensure the policy group is applied to each constituent individually. A shared QoS policy group will enforce the ceiling for the total throughput of all workloads.

Volume provisioning examples

Here's an example with defaults defined:

```
---
version: 1
storageDriverName: ontap-nas
backendName: customBackendName
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
labels:
  k8scluster: dev1
  backend: dev1-nasbackend
svm: trident_svm
username: cluster-admin
password: password
limitAggregateUsage: 80%
limitVolumeSize: 50Gi
nfsMountOptions: nfsvers=4
debugTraceFlags:
  api: false
  method: true
defaults:
  spaceReserve: volume
  qosPolicy: premium
  exportPolicy: myk8scluster
  snapshotPolicy: default
  snapshotReserve: '10'
```

For `ontap-nas` and `ontap-nas-flexgroups`, Astra Trident now uses a new calculation to ensure that the FlexVol is sized correctly with the snapshotReserve percentage and PVC. When the user requests a PVC, Astra Trident creates the original FlexVol with more space by using the new calculation. This calculation ensures that the user receives the writable space they requested for in the PVC, and not lesser space than what they requested. Before v21.07, when the user requests a PVC (for example, 5GiB), with the snapshotReserve to 50 percent, they get only 2.5GiB of writeable space. This is because what the user requested for is the whole volume and `snapshotReserve` is a percentage of that. With Trident 21.07, what the user requests for is the writeable space and Astra Trident defines the `snapshotReserve` number as the percentage of the whole volume. This does not apply to `ontap-nas-economy`. See the following example to see how this works:

The calculation is as follows:

```
Total volume size = (PVC requested size) / (1 - (snapshotReserve
percentage) / 100)
```

For snapshotReserve = 50%, and PVC request = 5GiB, the total volume size is 2/.5 = 10GiB and the available size is 5GiB, which is what the user requested in the PVC request. The `volume show` command should show results similar to this example:

```
Vserver   Volume       Aggregate    State      Type       Size  Available Used%
--------- ------------ ------------ ---------- ---- ---------- ---------- -----
                  _pvc_89f1c156_3801_4de4_9f9d_034d54c395f4
                                   online     RW         10GB     5.00GB    0%
                  _pvc_e8372153_9ad9_474a_951a_08ae15e1c0ba
                                   online     RW          1GB    511.8MB    0%
2 entries were displayed.
```

Existing backends from previous installs will provision volumes as explained above when upgrading Astra Trident. For volumes that you created before upgrading, you should resize their volumes for the change to be observed. For example, a 2GiB PVC with `snapshotReserve=50` earlier resulted in a volume that provides 1GiB of writable space. Resizing the volume to 3GiB, for example, provides the application with 3GiB of writable space on a 6 GiB volume.

Examples

Minimal configuration examples

The following examples show basic configurations that leave most parameters to default. This is the easiest way to define a backend.

> If you are using Amazon FSx on NetApp ONTAP with Trident, the recommendation is to specify DNS names for LIFs instead of IP addresses.

Default options on `ontap-nas-economy`

```
---
version: 1
storageDriverName: ontap-nas-economy
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
svm: svm_nfs
username: vsadmin
password: password
```

Certificate-based authentication

This is a minimal backend configuration example. `clientCertificate`, `clientPrivateKey`, and `trustedCACertificate` (optional, if using trusted CA) are populated in `backend.json` and take the base64-encoded values of the client certificate, private key, and trusted CA certificate, respectively.

```
---
version: 1
backendName: DefaultNASBackend
storageDriverName: ontap-nas
managementLIF: 10.0.0.1
dataLIF: 10.0.0.15
svm: nfs_svm
clientCertificate: ZXR0ZXJwYXB...ICMgJ3BhcGVyc2
clientPrivateKey: vciwKIyAgZG...0cnksIGRlc2NyaX
trustedCACertificate: zcyBbaG...b3Igb3duIGNsYXNz
storagePrefix: myPrefix_
```

Auto export policy

These examples show you how you can instruct Astra Trident to use dynamic export policies to create and manage the export policy automatically. This works the same for the `ontap-nas-economy` and `ontap-nas-flexgroup` drivers.

ontap-nas driver

```
---
version: 1
storageDriverName: ontap-nas
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
svm: svm_nfs
labels:
  k8scluster: test-cluster-east-1a
  backend: test1-nasbackend
autoExportPolicy: true
autoExportCIDRs:
- 10.0.0.0/24
username: admin
password: password
nfsMountOptions: nfsvers=4
```

`ontap-nas-flexgroup` **driver**

```
---
version: 1
storageDriverName: ontap-nas-flexgroup
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
labels:
  k8scluster: test-cluster-east-1b
  backend: test1-ontap-cluster
svm: svm_nfs
username: vsadmin
password: password
```

Using IPv6 addresses

This example shows `managementLIF` using an IPv6 address.

```
---
version: 1
storageDriverName: ontap-nas
backendName: nas_ipv6_backend
managementLIF: "[5c5d:5edf:8f:7657:bef8:109b:1b41:d491]"
labels:
  k8scluster: test-cluster-east-1a
  backend: test1-ontap-ipv6
svm: nas_ipv6_svm
username: vsadmin
password: password
```

`ontap-nas-economy` driver

```
---
version: 1
storageDriverName: ontap-nas-economy
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
svm: svm_nfs
username: vsadmin
password: password
```

`ontap-nas` driver for Amazon FSx for ONTAP using SMB volumes

```
---
version: 1
backendName: SMBBackend
storageDriverName: ontap-nas
managementLIF: example.mgmt.fqdn.aws.com
nasType: smb
dataLIF: 10.0.0.15
svm: nfs_svm
clientCertificate: ZXR0ZXJwYXB...ICMgJ3BhcGVyc2
clientPrivateKey: vciwKIyAgZG...0cnksIGRlc2NyaX
trustedCACertificate: zcyBbaG...b3Igb3duIGNsYXNz
storagePrefix: myPrefix_
```

Examples of backends with virtual pools

In the sample backend definition file shown below, specific defaults are set for all storage pools, such as `spaceReserve` at none, `spaceAllocation` at false, and `encryption` at false. The virtual pools are defined in the storage section.

Astra Trident sets provisioning labels in the “Comments” field. Comments are set on FlexVol for `ontap-nas` or FlexGroup for `ontap-nas-flexgroup`. Astra Trident copies all labels present on a virtual pool to the storage volume at provisioning. For convenience, storage administrators can define labels per virtual pool and group volumes by label.

In this example, some of the storage pool sets their own `spaceReserve`, `spaceAllocation`, and `encryption` values, and some pools overwrite the default values set above.

`ontap-nas` **driver**

```
---
version: 1
storageDriverName: ontap-nas
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
svm: svm_nfs
username: admin
password: password
nfsMountOptions: nfsvers=4
defaults:
  spaceReserve: none
  encryption: 'false'
  qosPolicy: standard
labels:
  store: nas_store
  k8scluster: prod-cluster-1
region: us_east_1
storage:
- labels:
    app: msoffice
    cost: '100'
  zone: us_east_1a
  defaults:
    spaceReserve: volume
    encryption: 'true'
    unixPermissions: '0755'
    adaptiveQosPolicy: adaptive-premium
- labels:
    app: slack
    cost: '75'
  zone: us_east_1b
  defaults:
    spaceReserve: none
    encryption: 'true'
    unixPermissions: '0755'
- labels:
    app: wordpress
    cost: '50'
  zone: us_east_1c
  defaults:
    spaceReserve: none
    encryption: 'true'
    unixPermissions: '0775'
```

```
- labels:
    app: mysqldb
    cost: '25'
  zone: us_east_1d
  defaults:
    spaceReserve: volume
    encryption: 'false'
    unixPermissions: '0775'
```

`ontap-nas-flexgroup` **driver**

```
---
version: 1
storageDriverName: ontap-nas-flexgroup
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
svm: svm_nfs
username: vsadmin
password: password
defaults:
  spaceReserve: none
  encryption: 'false'
labels:
  store: flexgroup_store
  k8scluster: prod-cluster-1
region: us_east_1
storage:
- labels:
    protection: gold
    creditpoints: '50000'
  zone: us_east_1a
  defaults:
    spaceReserve: volume
    encryption: 'true'
    unixPermissions: '0755'
- labels:
    protection: gold
    creditpoints: '30000'
  zone: us_east_1b
  defaults:
    spaceReserve: none
    encryption: 'true'
    unixPermissions: '0755'
- labels:
    protection: silver
    creditpoints: '20000'
  zone: us_east_1c
  defaults:
    spaceReserve: none
    encryption: 'true'
    unixPermissions: '0775'
- labels:
    protection: bronze
    creditpoints: '10000'
```

```
  zone: us_east_1d
  defaults:
    spaceReserve: volume
    encryption: 'false'
    unixPermissions: '0775'
```

`ontap-nas-economy` **driver**

```
---
version: 1
storageDriverName: ontap-nas-economy
managementLIF: 10.0.0.1
dataLIF: 10.0.0.2
svm: svm_nfs
username: vsadmin
password: password
defaults:
  spaceReserve: none
  encryption: 'false'
labels:
  store: nas_economy_store
region: us_east_1
storage:
- labels:
    department: finance
    creditpoints: '6000'
  zone: us_east_1a
  defaults:
    spaceReserve: volume
    encryption: 'true'
    unixPermissions: '0755'
- labels:
    department: legal
    creditpoints: '5000'
  zone: us_east_1b
  defaults:
    spaceReserve: none
    encryption: 'true'
    unixPermissions: '0755'
- labels:
    department: engineering
    creditpoints: '3000'
  zone: us_east_1c
  defaults:
    spaceReserve: none
    encryption: 'true'
    unixPermissions: '0775'
- labels:
    department: humanresource
    creditpoints: '2000'
  zone: us_east_1d
```

```
  defaults:
    spaceReserve: volume
    encryption: 'false'
    unixPermissions: '0775'
```

Update `dataLIF` after initial configuration

You can change the data LIF after initial configuration by running the following command to provide the new backend JSON file with updated data LIF.

```
tridentctl update backend <backend-name> -f <path-to-backend-json-file-with-updated-dataLIF>
```

> If PVCs are attached to one or multiple pods, you must bring down all corresponding pods and then bring them back up in order to for the new data LIF to take effect.

Map backends to StorageClasses

The following StorageClass definitions refer to the above virtual pools. Using the `parameters.selector` field, each StorageClass calls out which virtual pool(s) can be used to host a volume. The volume will have the aspects defined in the chosen virtual pool.

- The first StorageClass (`protection-gold`) will map to the first, second virtual pool in the `ontap-nas-flexgroup` backend and the first virtual pool in the `ontap-san` backend. These are the only pool offering gold level protection.
- The second StorageClass (`protection-not-gold`) will map to the third, fourth virtual pool in `ontap-nas-flexgroup` backend and the second, third virtual pool in `ontap-san` backend. These are the only pools offering protection level other than gold.
- The third StorageClass (`app-mysqldb`) will map to the fourth virtual pool in `ontap-nas` backend and the third virtual pool in `ontap-san-economy` backend. These are the only pools offering storage pool configuration for mysqldb type app.
- The fourth StorageClass (`protection-silver-creditpoints-20k`) will map to the third virtual pool in `ontap-nas-flexgroup` backend and the second virtual pool in `ontap-san` backend. These are the only pools offering gold-level protection at 20000 creditpoints.
- The fifth StorageClass (`creditpoints-5k`) will map to the second virtual pool in `ontap-nas-economy` backend and the third virtual pool in `ontap-san` backend. These are the only pool offerings at 5000 creditpoints.

Astra Trident will decide which virtual pool is selected and will ensure the storage requirement is met.

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: protection-gold
provisioner: netapp.io/trident
parameters:
  selector: "protection=gold"
  fsType: "ext4"
---
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: protection-not-gold
provisioner: netapp.io/trident
parameters:
  selector: "protection!=gold"
  fsType: "ext4"
---
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: app-mysqldb
provisioner: netapp.io/trident
parameters:
  selector: "app=mysqldb"
  fsType: "ext4"
---
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: protection-silver-creditpoints-20k
provisioner: netapp.io/trident
parameters:
  selector: "protection=silver; creditpoints=20000"
  fsType: "ext4"
---
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: creditpoints-5k
provisioner: netapp.io/trident
parameters:
  selector: "creditpoints=5000"
  fsType: "ext4"
```

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